Environment-Assisted Quantum Transport in Open Quantum Chains: Optimal Dephasing, Disorder Statistics, and the Limits of the Enhancement Ratio

A. Smith*

Ember Professional Research Division

(Dated: October 27, 2026)

Abstract

We study environment-assisted quantum transport (ENAQT) in tight-binding chains with site-local pure dephasing, an irreversible trap and uniform excitation loss, and compute the trapping yield $\eta(\gamma_\phi)$ exactly from one linear solve per parameter point. For linear energy funnels of $N = 2$–20 sites at fixed total bias, the optimal dephasing rate equals the nearest-neighbour energy mismatch, $\gamma_\phi^* = \varepsilon_{\mathrm{tot}}/(N-1)$, to within 2%, the trapped-chain counterpart of the rule found for boundary-driven tilted chains; it maximizes the Haken–Strobl–Reineker hopping rate on every bond at once and holds to within 15% over two decades of trap and loss rates. Because the dephasing acts at infinite temperature, the yield is exactly invariant under reversing the bias, and a flat chain, which shows no ENAQT, reaches a higher yield than any funnel: dephasing only repairs yield lost to energy mismatch. The enhancement ratio $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}$ is set by its reference point and by the rates: for $\varepsilon_{\mathrm{tot}} = 5\Delta$, above the bandwidth, it plateaus at 36–38 against $\gamma_\phi = 10^{-3}\Delta$ but reaches 1.3×10^3 at $N = 20$ against the coherent limit, whereas for $\varepsilon_{\mathrm{tot}} = 2\Delta$ it stays near 1.6. In 1,000 chains with Gaussian site disorder ($\sigma = 2\Delta$), 97.4% show ENAQT and median ratios exceed the funnel's because the coherent baseline collapses; only 4 of these chains beat the flat chain, while weaker disorder of matched nearest-neighbour strength beats the funnel. Maximizing the ratio over site energies has no finite optimum: it drives the profile to the bounds of the search space and lowers the yield. For the seven-site FMO complex with the trap at BChl 3, $\gamma_\phi^* = 1.9\,\Delta_{\mathrm{FMO}}$, of the order of a high-temperature estimate of the protein dephasing rate, while the 4.7-fold enhancement falls to 1.14 at biological trap and loss rates. Peak yield and optimal dephasing rate, not the enhancement ratio, are the robust figures of merit.

I. INTRODUCTION

Environment-assisted quantum transport (ENAQT), also called dephasing-assisted or noise-assisted transport, is the observation that moderate environmental dephasing can increase the probability that an excitation reaches a trap in a network of coupled sites [1–4]. At weak dephasing, transport is limited by coherent effects: energy mismatch between sites, destructive interference and Anderson localization [5]. At strong dephasing, frequent ef-

* grandmasta1024@gmail.com

fective measurement of site populations freezes the dynamics (a quantum Zeno effect). In between, dephasing broadens site energies enough to bridge mismatches without freezing the motion, and the trapping yield $\eta(\gamma_\phi)$ has an interior maximum. In the incoherent limit this is the classic Haken–Strobl–Reineker (HSR) picture of exciton hopping in a dephasing environment [6–8].

The effect has been analyzed in light-harvesting models such as the Fenna–Matthews–Olson (FMO) complex [1–4, 9, 10], in ordered systems [11], in disordered chains [8, 12–16] and networks [17, 18], in tilted (Wannier–Stark) chains [19, 20], in lossy lattices [21] and in strongly correlated boundary-driven systems [22]. It has been observed in an optical cavity network [23], a photonic waveguide network [24], superconducting circuits [25], a ten-qubit trapped-ion network [26] and a photonic simulation of the FMO complex [27]. Coates, Lovett and Gauger showed that disordered networks can have more than one optimal dephasing regime [18]; Zerah-Harush and Dubi traced ENAQT to a competition between dephasing-driven population equalization and the source–sink gradient [17] and studied how disorder and interactions modify it [15]; Jacob *et al.* showed that in a boundary-driven tilted chain the current peaks when the dephasing rate equals the nearest-neighbour energy difference, the Bloch-oscillation frequency, so that at fixed total tilt the optimum falls as $1/L$ with chain length [20]. Length laws for the optimal dephasing rate have also been found in disordered chains with a sink [14], where it is tied to eigenstate localization [16], and in lossy lattices [21]. Recent work optimizes site energies [28] and site-dependent dephasing [29] for the absolute steady-state flux, and an evidence review of quantum effects in biology treats noise-assisted transport as a conditional design principle, to be tested by dephasing perturbations linked to functional yield [30].

A quantity that recurs throughout this literature, and in our own earlier drafts, is the enhancement ratio $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}$: the peak yield divided by a reference yield at weak or vanishing dephasing. That such enhancements depend on the comparison regime has been noted before [31], very large ratios have been reported where the undephased transport is already tiny [32], and at realistic parameters the gain in photosynthetic complexes appears to be small [33]. Using a minimal and fully reproducible Lindblad model, we separate what is robust from what is not. (i) In linear energy funnels the optimum of the trapping yield is fixed by the nearest-neighbour mismatch, $\gamma_\phi^* = \varepsilon_{\mathrm{tot}}/(N-1)$, the HSR resonance condition applied to every bond and the trapped-chain counterpart of the boundary-driven result of

Ref. [20]; this rule survives large changes of the trap and loss rates. (ii) The yield is exactly symmetric under reversal of the bias, and the flat chain outperforms every funnel; dephasing only recovers what the mismatch costs. (iii) The enhancement ratio has no unique length dependence: its apparent saturation in funnels depends on the reference rate, the total bias and the loss rate, and its large values in disordered chains, or in landscapes optimized for it, come from a collapsing coherent baseline rather than from better transport. We close with a seven-site FMO benchmark, including biological trap and loss rates, and a first look at square lattices.

II. MODEL AND METHOD

A. Master equation

For N sites $|j\rangle$, $j = 1, \ldots, N$, with Hamiltonian H, the density matrix obeys

$$\dot{\rho} = -i[H, \rho] + \gamma_\phi \sum_j \left(P_j \rho P_j - \tfrac{1}{2}\{P_j, \rho\} \right) - \frac{\kappa}{2}\{P_N, \rho\} - \Gamma\rho, \tag{1}$$

with $P_j = |j\rangle\langle j|$ and $\hbar = 1$. The second term is site-local pure dephasing, which damps every coherence ρ_{mn} $(m \neq n)$ at rate γ_ϕ and leaves populations unchanged; it is the HSR model at infinite temperature [6, 7]. The third term is irreversible trapping from site N at rate κ, and the last term is a uniform loss (recombination or fluorescence) at rate Γ. The excitation starts on site 1, $\rho(0) = P_1$.

B. Trapping yield

The yield is the probability that the excitation is trapped before it is lost,

$$\eta = \kappa \int_0^\infty \rho_{NN}(t)\, dt. \tag{2}$$

With column-stacked vec(ρ), Eq. (1) reads $\partial_t \mathrm{vec}(\rho) = \mathcal{L}\,\mathrm{vec}(\rho)$ with

$$\begin{aligned} \mathcal{L} = & -i\left(\mathbf{1} \otimes H - H^T \otimes \mathbf{1}\right) + \gamma_\phi \sum_j \left(P_j \otimes P_j - \tfrac{1}{2}\mathbf{1} \otimes P_j - \tfrac{1}{2} P_j \otimes \mathbf{1}\right) \\ & - \frac{\kappa}{2}\left(\mathbf{1} \otimes P_N + P_N \otimes \mathbf{1}\right) - \Gamma\, \mathbf{1}_{N^2}. \end{aligned} \tag{3}$$

For $\Gamma > 0$ every eigenvalue of $\mathcal{L}$ has a negative real part, so

$$\eta = \kappa \left[-\mathcal{L}^{-1}\mathrm{vec}(\rho(0))\right]_{NN}, \tag{4}$$

one $N^2 \times N^2$ linear solve per parameter point. Writing $\mathcal{L} = \mathcal{L}_{\mathrm{base}} + \gamma_\phi \mathcal{L}_{\mathrm{deph}}$, a sweep over γ_ϕ reuses both matrices.

C. Two exact properties

Bias reversal. For a real Hamiltonian on a bipartite lattice (a chain or a square lattice), the yield is invariant under $\varepsilon_j \to -\varepsilon_j$ for all sites at once. Complex conjugation of Eq. (1) maps the evolution under H onto the evolution under $-H$, since the dephasing, trap and loss terms are real, and the gauge transformation $|j\rangle \to (-1)^j |j\rangle$ restores the sign of the couplings. Populations, and hence η, are unchanged by both operations. An uphill funnel and the corresponding downhill funnel therefore have identical yields at every γ_ϕ: infinite-temperature dephasing provides no downhill drift. The analogous statement for the boundary-driven current of a tilted chain is noted in Refs. [20, 34], and under HSR dephasing forward and backward hopping rates are equal [8].

Incoherent hopping rate. For two sites with mismatch $\delta\varepsilon$ and coupling J, adiabatic elimination of the coherence in Eq. (1) gives the HSR rate [7, 8]

$$k(\gamma_\phi) = \frac{2J^2\gamma_\phi}{\gamma_\phi^2 + \delta\varepsilon^2}, \tag{5}$$

symmetric in $\delta\varepsilon$ and maximal at $\gamma_\phi = |\delta\varepsilon|$, where $k = J^2/|\delta\varepsilon|$. The elimination is controlled for $J \ll |\gamma_\phi + i\delta\varepsilon|$; we use Eq. (5) only to interpret the exact results.

D. Hamiltonians and parameters

Chains. Nearest-neighbour coupling $\Delta = 1$ sets the energy and inverse-time unit (bandwidth 4Δ). The linear energy funnel has site energies $\varepsilon_j = \varepsilon_{\mathrm{tot}}/2 - (j-1)\varepsilon_{\mathrm{tot}}/(N-1)$ with $\varepsilon_{\mathrm{tot}} = 5\Delta$ unless stated otherwise, so the total donor–trap bias is fixed as N grows and the nearest-neighbour mismatch is $\delta\varepsilon = \varepsilon_{\mathrm{tot}}/(N-1)$. Disordered chains draw $\varepsilon_j \sim \mathcal{N}(0, \sigma^2)$ independently.

Two-site model. For $N = 2$ we use the spin-boson convention $H = \frac{\varepsilon}{2}\sigma_z + \frac{\Delta}{2}\sigma_x$, i.e. an inter-site coupling of $\Delta/2$; this is the convention used in Sec. III A and Fig. 1. The $N = 2$

funnel of Sec. III B uses coupling Δ and therefore gives a different enhancement (2.75 rather than 7.20 at $\varepsilon = 5\Delta$).

Rates and grids. Unless stated otherwise, $\kappa = 0.1\Delta$ and $\Gamma = 0.01\Delta$. One-dimensional sweeps use log-spaced $\gamma_\phi \in [10^{-3}, 10^{3}]\Delta$ with 300 points for ordered chains (grid step 4.7%) and 100 points for disorder ensembles (step 15%). For the funnels we also refine γ_ϕ^* with a bounded scalar optimizer; the refined peak yields differ from the grid values by less than 10^{-4} (relative).

Enhancement ratio. We define $\eta_{\mathrm{peak}} = \max_{\gamma_\phi} \eta(\gamma_\phi)$ and γ_ϕ^* its location. The reference yield is $\eta_{\mathrm{ref}} = \eta(10^{-3}\Delta)$, the lowest grid point, unless explicitly marked as the true coherent limit $\eta(0)$; Sec. III B shows that this choice matters. A realization is counted as showing ENAQT when γ_ϕ^* is an interior grid point and $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}} > 1.01$.

III. RESULTS

A. Two sites: the role of bias and trap rate

Figure 1(a) and Table I show the two-site yield. Without bias the yield decreases monotonically with γ_ϕ: coherent transfer between degenerate sites is already efficient and dephasing only slows it. With bias, the coherent yield drops (the excitation oscillates with reduced amplitude $\propto \Delta^2/(\Delta^2 + \varepsilon^2)$) while dephasing-assisted hopping restores it, and the interior maximum sits close to the HSR resonance $\gamma_\phi = \varepsilon$ of Eq. (5): the refined ratio $\gamma_\phi^*/\varepsilon$ rises from 0.88 at $\varepsilon = 0.5\Delta$ to 0.99 at $\varepsilon = 5\Delta$, approaching one as the bias grows relative to the coupling, as in the two-site result of Ref. [20] and the line-broadening picture of Ref. [2]. At very strong dephasing all curves collapse onto the Zeno tail, $\eta(10^3\Delta) = 0.043$.

The trap rate sets how strongly the ratio depends on the baseline (Fig. 1(b), Table II). A slow trap ($\kappa = 0.01\Delta$) gives the largest ratio (19.3) but the lowest yield (0.29); a fast trap ($\kappa = \Delta$) gives the highest yield (0.82) but a ratio of only 1.65, because the coherent baseline is already large. The peak-to-Zeno ratio varies much less (12.6–17.4).

B. Linear energy funnels

Figure 2 and Table III summarize funnels of $N = 2$–20 sites at fixed total bias $\varepsilon_{\mathrm{tot}} = 5\Delta$. Every funnel shows an interior yield maximum.

TABLE I. Two-site model ($H = \frac{\varepsilon}{2}\sigma_z + \frac{\Delta}{2}\sigma_x$, $\kappa = 0.1\Delta$, $\Gamma = 0.01\Delta$). The enhancement is given against both references.

ε [Δ]	γ_ϕ^* [Δ]	$\eta_{\rm peak}$	$\eta(\gamma_\phi = 0)$	$\eta_{\rm peak}/\eta(10^{-3}\Delta)$	$\eta_{\rm peak}/\eta(0)$
0.0	—	0.832	0.832	1.00	1.00
0.5	0.45	0.818	0.773	1.06	1.06
1.0	0.93	0.804	0.638	1.26	1.26
2.0	1.95	0.776	0.375	2.05	2.07
5.0	4.92	0.704	0.096	7.20	7.30

TABLE II. Dependence on the trap rate (two-site model, $\varepsilon = 5\Delta$, $\Gamma = 0.01\Delta$). $\eta_{\rm Zeno} = \eta(\gamma_\phi = 10^3\Delta)$.

κ [Δ]	γ_ϕ^* [Δ]	$\eta_{\rm peak}$	$\eta_{\rm peak}/\eta_{\rm ref}$	$\eta_{\rm peak}/\eta_{\rm Zeno}$
0.01	4.92	0.294	19.3	12.6
0.1	4.92	0.704	7.20	16.3
1.0	4.49	0.818	1.65	17.4

The mismatch rule. The optimal dephasing rate equals the nearest-neighbour mismatch,

$$\gamma_\phi^* \simeq \delta\varepsilon = \frac{\varepsilon_{\rm tot}}{N-1}, \tag{6}$$

with the refined $\gamma_\phi^*/\delta\varepsilon$ between 0.988 and 1.018 for $N = 2$–20 (grid values 0.975–1.032; Fig. 2(c)). Because every bond of a linear funnel has the same mismatch, Eq. (5) is maximized on all bonds simultaneously at $\gamma_\phi = \delta\varepsilon$, which is also the Bloch frequency of the tilted chain. This is the rule derived by Jacob *et al.* for the boundary-driven current, $\gamma_{\rm max} \simeq U$ with U the nearest-neighbour energy difference [20], here confirmed quantitatively for the trapping yield of chains with uniform loss; Moix *et al.* used the same Lorentzian rate to predict an optimal dephasing equal to the disorder width for diffusion in disordered chains [8]. A power-law fit over the same range gives $\gamma_\phi^* \approx 9.2\,\Delta\,N^{-1.20}$ ($R^2 = 0.992$), but this exponent only describes the curvature of $\varepsilon_{\rm tot}/(N-1)$ on a log-log plot over $N = 2$–20 (a power fit of the mismatch itself gives -1.22); it carries no separate physics.

Peak yield, bias symmetry and the flat chain. The peak yield falls steadily, from 0.80 at $N = 2$ to 0.25 at $N = 20$, because the transit time grows with N while the loss rate Γ is

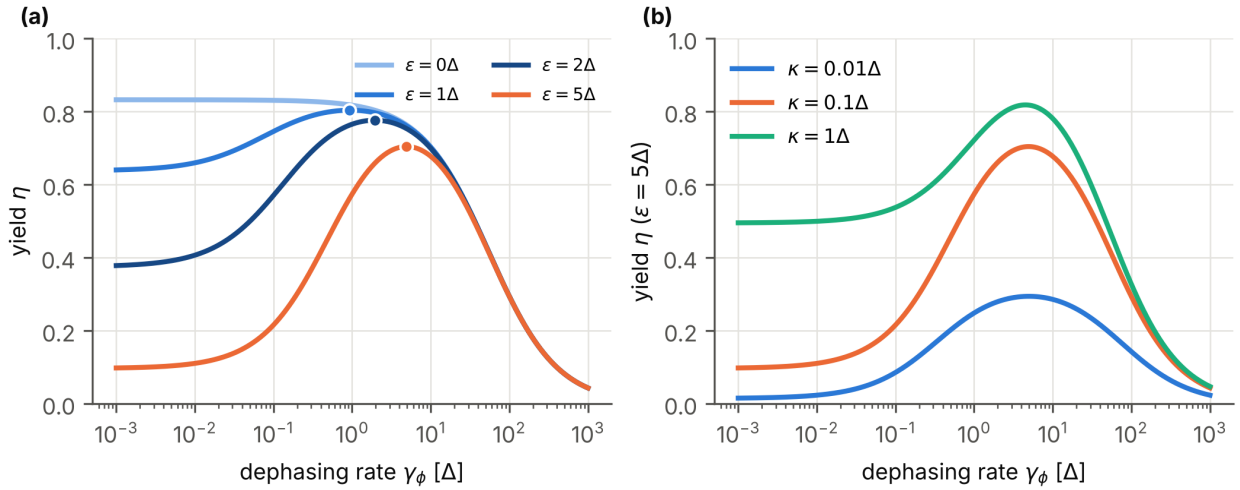


FIG. 1. Two-site yield. (a) $\eta(\gamma_\phi)$ for biases $\varepsilon = 0, 1, 2, 5\Delta$ ($\kappa = 0.1\Delta$); dots mark interior maxima. (b) $\eta(\gamma_\phi)$ at $\varepsilon = 5\Delta$ for three trap rates.

fixed. Reversing the funnel into an uphill ramp leaves every yield unchanged, as required by Sec. II C. The flat chain ($\varepsilon_j = 0$) shows no ENAQT: its yield decreases monotonically with γ_ϕ at the default rates and over $\kappa = 0.01$–$100\,\Delta$ and $\Gamma = 10^{-3}$–$10^{-1}\Delta$ for $N = 3$–7. This is the end-to-end geometry, with the excitation starting at the end opposite the trap, for which dephasing was found numerically not to help by Plenio and Huelga [2] and explained by Kassal and Aspuru-Guzik in the weak trapping and loss limit [11]; in other source–trap geometries ordered systems can show ENAQT [11]. Its coherent yield is nevertheless higher than the funnel's peak yield at every length we checked: 0.773 versus 0.722 at $N = 3$, 0.625 versus 0.518 at $N = 7$, and 0.461 versus 0.320 at $N = 15$. In this model, then, dephasing does not give the funnel a transport advantage; it recovers part of the yield that the energy mismatch removes. A downhill preference would require thermal relaxation, which pure dephasing does not contain.

The enhancement ratio. Measured against $\gamma_\phi = 10^{-3}\Delta$ the ratio grows almost linearly for $N \leq 10$,

$$\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}} \approx 3.85\,N - 4.9 \qquad (N \leq 10,\ R^2 = 0.996), \tag{7}$$

then plateaus at 36–38 for $N = 12$–20. Measured against the true coherent limit, the same data give $\eta_{\mathrm{peak}}/\eta(0) = 107$ at $N = 10$, 407 at $N = 15$ and 1.3×10^3 at $N = 20$ (Fig. 2(b)). The plateau therefore does not reflect a limit on noise-assisted transport; it appears because, in long funnels, even $\gamma_\phi = 10^{-3}\Delta$ already lifts the yield well above its coherent value, by a

factor of 3.3 at $N = 10$, 10.7 at $N = 15$ and 35 at $N = 20$.

TABLE III. Linear energy funnels ($\varepsilon_{\rm tot} = 5\Delta$, $\kappa = 0.1\Delta$, $\Gamma = 0.01\Delta$). γ_ϕ^* is the grid optimum and $\gamma_\phi^*/\delta\varepsilon$ the refined optimum divided by the nearest-neighbour mismatch $\delta\varepsilon = \varepsilon_{\rm tot}/(N-1)$. Two enhancement ratios are given: against $\eta(10^{-3}\Delta)$ and against the coherent limit $\eta(0)$.

N	γ_ϕ^* [Δ]	$\gamma_\phi^*/\delta\varepsilon$	$\eta_{\rm peak}$	$\eta(0)$	$\eta_{\rm peak}/\eta(10^{-3}\Delta)$	$\eta_{\rm peak}/\eta(0)$
2	4.92	0.988	0.797	0.286	2.75	2.78
3	2.46	0.988	0.722	0.105	6.27	6.84
5	1.23	0.992	0.605	0.032	14.3	19.0
7	0.81	0.997	0.518	0.013	22.8	41.4
10	0.56	1.004	0.424	3.9×10^{-3}	32.4	107
15	0.35	1.012	0.320	7.9×10^{-4}	37.9	407
20	0.27	1.018	0.254	1.9×10^{-4}	37.3	1314

Robustness. We repeated the funnel scan on a 3×3 grid of trap and loss rates, $\kappa \in \{0.01, 0.1, 1\}\Delta$ and $\Gamma \in \{10^{-3}, 10^{-2}, 10^{-1}\}\Delta$. The mismatch rule survives with $\gamma_\phi^*/\delta\varepsilon = 0.85$–$1.09$ over all nine combinations; the largest deviations occur for the fast trap $\kappa = \Delta$, which broadens the last site and shifts γ_ϕ^* downward. The enhancement ratio does not: at $N = 20$ it ranges from 4.7 ($\kappa = \Delta$, $\Gamma = 10^{-3}\Delta$) to 91 ($\kappa = 0.01\Delta$, $\Gamma = 0.1\Delta$) against $\gamma_\phi = 10^{-3}\Delta$. The plateau also depends on the total bias. For $\varepsilon_{\rm tot} = 2\Delta$, below the bandwidth 4Δ, the ratio is only 1.53–1.66 for $N = 5$–20 (1.60–1.73 against $\eta(0)$), whereas for $\varepsilon_{\rm tot} = 8\Delta$ it reaches 843 against $10^{-3}\Delta$ and 6.3×10^8 against $\eta(0)$ at $N = 20$. A natural reading is Wannier–Stark localization [35]: the eigenstates of a chain with field step $\delta\varepsilon$ extend over about $4\Delta/\delta\varepsilon = 4(N-1)\Delta/\varepsilon_{\rm tot}$ sites, which covers the whole chain only when $\varepsilon_{\rm tot} \lesssim 4\Delta$. Above that bias coherent transfer from site 1 to the trap is blocked and the baseline collapses; below it the baseline stays finite and the ratio is modest. The value 36–38 is therefore a property of $\varepsilon_{\rm tot} = 5\Delta$ and the default rates, not of funnels in general.

C. Random site disorder

We drew 100 disorder realizations with $\sigma = 2\Delta$ for each $N \in \{2, 3, 4, 5, 6, 7, 8, 10, 12, 15\}$, 1,000 chains in total; the same seeds (0–99) are used at every N. Table IV and Fig. 3 give

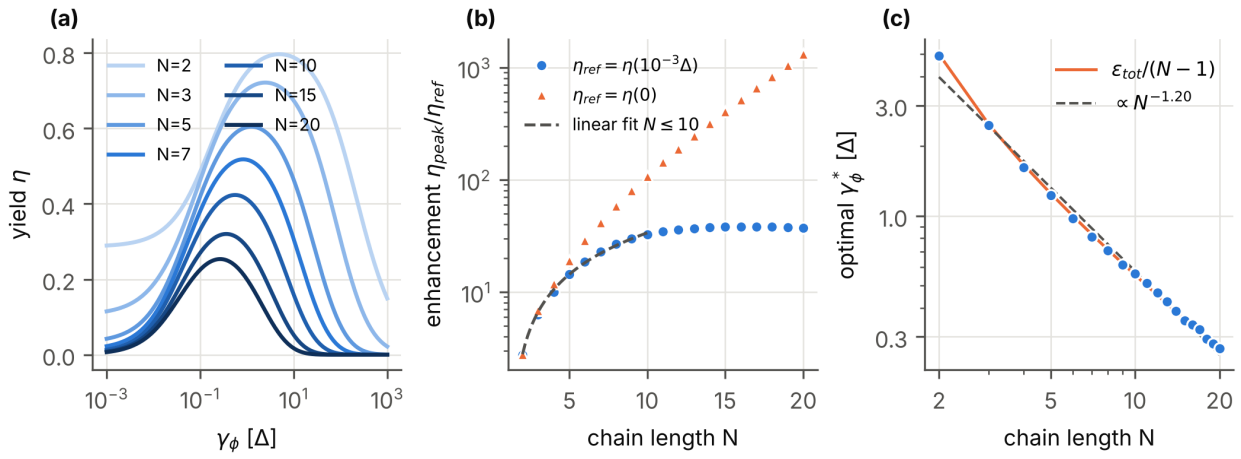


FIG. 2. Linear energy funnels ($\varepsilon_{\rm tot} = 5\Delta$). (a) Yield $\eta(\gamma_\phi)$ for $N = 2, 3, 5, 7, 10, 15, 20$. (b) Enhancement ratio against $\eta_{\rm ref} = \eta(10^{-3}\Delta)$ (circles; dashed line is the linear fit for $N \leq 10$) and against $\eta_{\rm ref} = \eta(0)$ (triangles). (c) Optimal dephasing rate with the nearest-neighbour mismatch $\varepsilon_{\rm tot}/(N-1)$ (solid) and the power-law fit (dashed).

the statistics; confidence intervals (CI) for medians are 95% bootstrap intervals from 2,000 resamples.

Prevalence. An interior maximum with at least 1% gain occurs in 974 of the 1,000 chains. The fraction is lowest for the shortest chains (89% at $N = 2$, 92% at $N = 3$), where a realization can have a small bias and behave like the degenerate case of Sec. III A, and is 98–100% for $N \geq 4$. So ENAQT is the typical behaviour of a disordered chain with a trap, not a fine-tuned one, in line with earlier studies [8, 13–16].

Ratio versus yield. The median ratio grows quickly with N and exceeds the ordered funnel's ratio (against $\gamma_\phi = 10^{-3}\Delta$) from $N = 10$ on: 35.7 (CI 21.6–48.1) versus 32.4 at $N = 10$, and 244 (CI 163–426) versus 37.9 at $N = 15$. The distribution is strongly right-skewed (mean 6.9×10^3 at $N = 15$, maximum 2.4×10^5). Figure 3(b) shows where the large ratios come from: the median peak yield decreases gently with N, from 0.82 to 0.13, while the median baseline collapses from 0.66 to 5.5×10^{-4}. This is the baseline effect behind the early observation that the average improvement is larger the more static disorder there is [3]. At this disorder strength the median peak yield is slightly above the funnel's for $N = 2$ and 3 (0.82 versus 0.80, 0.74 versus 0.72) and below it for $N \geq 4$ (0.43 versus 0.52 at $N = 7$; 0.13 versus 0.32 at $N = 15$). Measured against the true coherent limit the funnel's ratio is the larger one at $N = 10$ and 15 (107 versus a median of 41, and 407 versus 354). Against the

flat chain the comparison is one-sided: only 4 of the 1,000 disordered chains, all at $N = 3$, reach a higher peak yield than the flat chain of the same length.

Matched disorder. The comparison with the funnel depends on the disorder strength. Choosing $\sigma = \delta\varepsilon/\sqrt{2}$, so that the root-mean-square difference between neighbouring site energies equals the funnel's mismatch $\delta\varepsilon = 5\Delta/(N-1)$, the median peak yield (50 realizations per N) exceeds the funnel's at every length tested: 0.683 versus 0.658 at $N = 4$, 0.556 versus 0.518 at $N = 7$, 0.479 versus 0.424 at $N = 10$ and 0.418 versus 0.320 at $N = 15$. A random landscape with the same typical step as the funnel, but no net bias, transports better than the funnel, consistent with the bias symmetry of Sec. II C: the net drop of the funnel is a cost, not a help.

TABLE IV. Disorder ensemble ($\sigma = 2\Delta$, 100 realizations per N, $\kappa = 0.1\Delta$, $\Gamma = 0.01\Delta$). ENAQT: fraction with an interior maximum and $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}} > 1.01$. Ratios are against $\gamma_\phi = 10^{-3}\Delta$; IQR is the interquartile range. The funnel columns are the ordered funnel at the same N, computed as in Table III.

N	ENAQT	median $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}$ (IQR)	median η_{peak}	funnel $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}$	funnel η_{peak}
2	89%	1.23 (1.06–1.51)	0.82	2.75	0.80
3	92%	2.27 (1.15–3.75)	0.74	6.27	0.72
4	98%	3.55 (1.81–6.27)	0.65	10.0	0.66
5	100%	5.51 (2.39–13.0)	0.58	14.3	0.60
6	98%	9.88 (3.00–34.5)	0.50	18.7	0.56
7	100%	8.92 (3.71–49.1)	0.43	22.8	0.52
8	99%	13.8 (4.50–58.9)	0.38	26.6	0.48
10	100%	35.7 (8.96–132)	0.28	32.4	0.42
12	98%	72.3 (9.92–288)	0.22	35.9	0.38
15	100%	244 (47.9–1403)	0.13	37.9	0.32

Disorder strength. At $N = 7$ (50 realizations per σ) the median ratio is 1.00 at $\sigma = 0.25\Delta$, 1.74 at $\sigma = \Delta$, 8.3 at 2Δ, 97 at 3Δ and 7.2×10^3 at 5Δ, while the median peak yield falls from 0.60 to 0.26. For $\sigma \geq \Delta$ the median ratio is well described by $\ln(\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}) \approx 2.15\,\sigma/\Delta - 1.83$ ($R^2 = 0.996$; Fig. 3(c)). We do not attach a mechanism to this exponential form: it describes the range studied, and the baseline it reflects is set jointly by localization, the trap and the

loss rate. For orientation, the band-centre localization length of the infinite chain, from a transfer-matrix calculation with 4×10^5 sites, is $\xi = 139$ sites at $\sigma = 0.25\Delta$, 35 at 0.5Δ, 9.2 at Δ, 4.4 at 1.5Δ, 2.8 at 2Δ and 1.6 at 3Δ. At weak disorder this follows $\xi \approx 8.75\,\Delta^2/\sigma^2$, the Gaussian equivalent of the band-centre result $105\,\Delta^2/W^2$ for box disorder of width W [36]. The $N = 7$ chains are thus shorter than ξ for $\sigma \leq \Delta$ and several localization lengths long for $\sigma \geq 2\Delta$ ($7/\xi \approx 2.5$ at 2Δ).

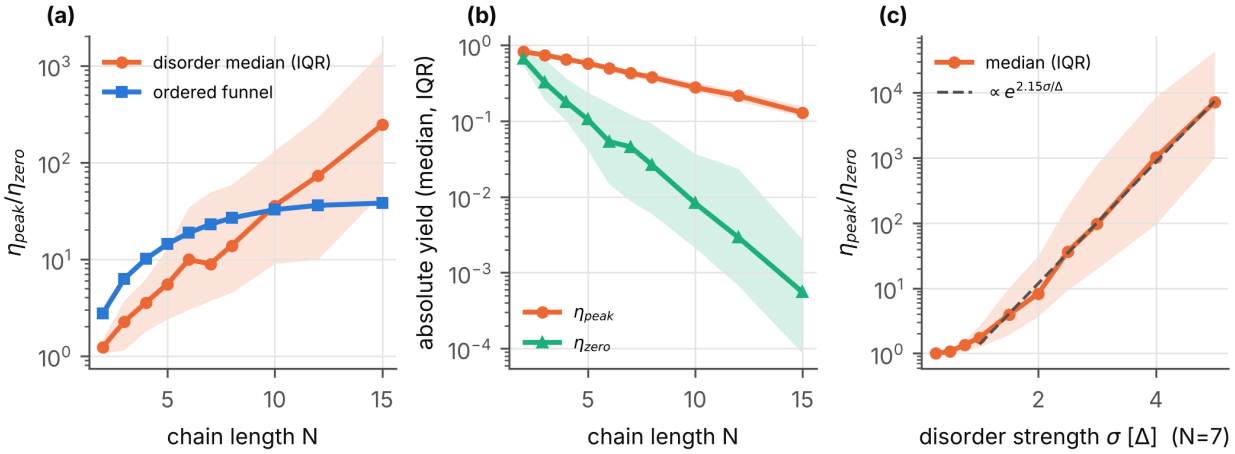


FIG. 3. Disorder ensemble. (a) Median enhancement ratio (shaded: interquartile range) for $\sigma = 2\Delta$ versus N, with the ordered funnel for comparison (both against $\gamma_\phi = 10^{-3}\Delta$). (b) Median peak yield η_{peak} and baseline η_{ref} for the same ensemble: the ratio grows because the baseline collapses. (c) Median ratio versus disorder strength at $N = 7$ with the exponential fit for $\sigma \geq \Delta$.

D. Optimizing the ratio

To see what the ratio rewards, we maximized $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}}$ over the site energies $\varepsilon_1, \ldots, \varepsilon_{N-1}$, each restricted to $[-5\Delta, 5\Delta]$, with $\varepsilon_N = 0$. We used the differential-evolution routine of SciPy (one run with seed 42, population $8(N-1)$, 100 generations, L-BFGS-B polish), with η_{peak} taken over 30 log-spaced points in $[10^{-3}, 10^{2}]\Delta$ and $\eta_{\mathrm{ref}} = \max[\eta(10^{-8}\Delta), 10^{-12}]$.

What the optimizer returns. For $N = 5$, 7 and 10 the result is a two-level step at the bounds: the first $\lfloor (N-1)/2 \rfloor$ sites at $+5\Delta$, the following $\lfloor (N-1)/2 \rfloor$ at -5Δ, and the trap at 0 (for $N = 10$ the ninth site sits at -0.28Δ; Fig. 4(a)). For $N = 3$ the profile is reversed, $(-5, +5, 0)\Delta$. For $N = 12$ and 15 the profiles are irregular; with a single run and no convergence check we treat them as not converged. These landscapes achieve very large

ratios (87 at $N = 3$, 9.0×10^3 at $N = 5$, 8.0×10^5 at $N = 7$) by making coherent transfer almost impossible.

The objective has no finite optimum. Fixing the step profile and varying its height B shows why the optimizer ends at the bounds (Table V): the ratio grows without limit as B increases, by orders of magnitude per doubling, while the peak yield falls. The returned profile is the corner of the search box, not a property of the transport problem.

TABLE V. Step profiles with height $\pm B$ (the first $\lfloor (N-1)/2 \rfloor$ sites at $+B$, the rest at $-B$, the trap at 0). The ratio is against $\eta(0)$.

	$N = 5$		$N = 7$	
B [Δ]	η_{peak}	$\eta_{\mathrm{peak}}/\eta(0)$	η_{peak}	$\eta_{\mathrm{peak}}/\eta(0)$
2.5	0.546	83	0.428	765
5	0.452	9.0×10^3	0.322	8.0×10^5
10	0.330	5.0×10^5	0.204	1.6×10^8
20	0.200	2.0×10^7	0.103	2.2×10^{10}

Comparison. The optimized peak yields are below those of simple landscapes that lie inside the same search box: 0.65, 0.32 and 0.21 at $N = 3$, 7 and 10, against 0.72, 0.52 and 0.42 for the $\varepsilon_{\mathrm{tot}} = 5\Delta$ funnel and 0.77, 0.62 and 0.55 for the flat chain. Figure 4(b) also shows a unit-step funnel, $\varepsilon_j = (N - j)\Delta$, used in earlier versions of this work; for $N > 6$ it extends beyond the box and has a larger total bias, so it is a reference curve rather than a competitor.

Numerical limits. The baseline proxy $\eta(10^{-8}\Delta)$ agrees with $\eta(0)$ to 0.2% for the optimized $N = 7$ profile but overestimates it by 16% at $N = 10$ (3.06×10^{-10} versus 2.64×10^{-10}), and for $N = 12$ and 15 the baseline approaches the 10^{-12} floor. Ratios for $N \geq 10$ in this section are therefore indicative only. Maximizing the enhancement ratio selects for a poor coherent baseline, not for good transport, and it should not be used as a design objective. Optimizing the absolute flux instead, as Lawrence *et al.* did with gradient methods [28], gives nearly flat landscapes for short-range tunnelling, consistent with the flat-chain result of Sec. III B.

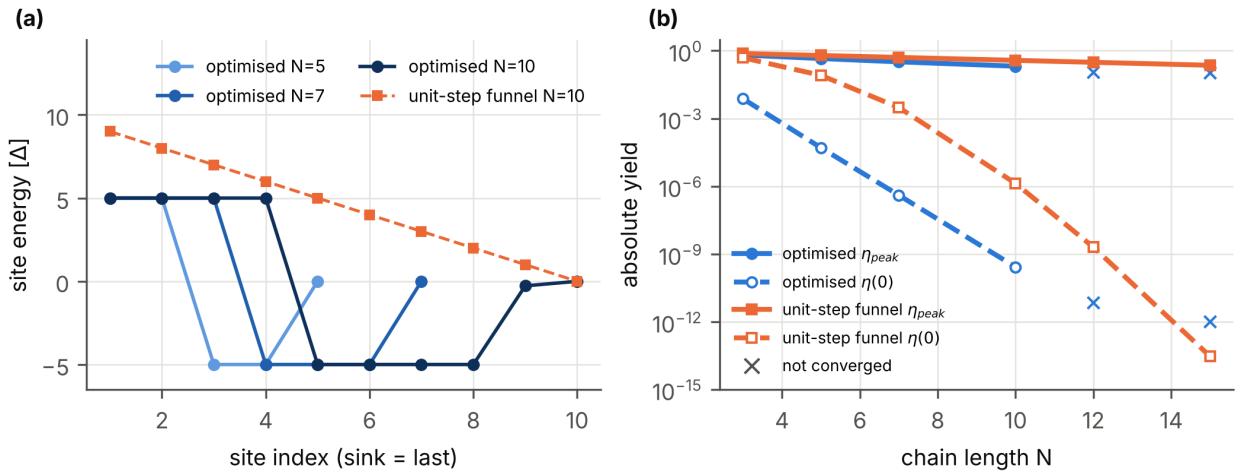


FIG. 4. Ratio-optimized site energies. (a) Optimized profiles for $N = 5, 7, 10$ and the unit-step funnel $\varepsilon_j = (N - j)\Delta$ for $N = 10$. (b) Peak yield (filled) and coherent yield $\eta(0)$ (open) for the optimized profiles and the unit-step funnel; crosses mark the non-converged optimizer runs ($N = 12$, 15).

E. FMO benchmark

For the seven-site FMO Hamiltonian of Adolphs and Renger [37] (site energies and couplings in cm^{-1}, expressed in units of $\Delta_{\mathrm{FMO}} = 87.7\,\mathrm{cm}^{-1}$, the BChl 1–2 coupling), with the excitation starting on BChl 1 and the trap on BChl 3, which faces the reaction center (Adolphs and Renger proposed BChls 3 and 4 as the linker pigments [37]; see also Ref. [3]), we find

$$\gamma_\phi^* = 1.87\,\Delta_{\mathrm{FMO}}, \qquad \eta_{\mathrm{peak}} = 0.46, \qquad \eta(0) = 0.098, \qquad \eta_{\mathrm{peak}}/\eta(0) = 4.7.$$

In time units, $\hbar/\Delta_{\mathrm{FMO}} = 60.5\,\mathrm{fs}$, so the default rates correspond to $\kappa^{-1} = 0.61\,\mathrm{ps}$ and $\Gamma^{-1} = 6.1\,\mathrm{ps}$.

Dephasing rate. For the Drude bath of Ishizaki and Fleming [38] ($\lambda = 35\,\mathrm{cm}^{-1}$, $\gamma_c^{-1} = 50\,\mathrm{fs}$, $T = 300\,\mathrm{K}$), the high-temperature pure-dephasing rate of a single site energy is $2\lambda k_B T/\hbar\gamma_c = 1.57\,\Delta_{\mathrm{FMO}}$. With independent baths on each site a site-to-site coherence decays at twice this rate, $4\lambda k_B T/\hbar\gamma_c = 3.1\,\Delta_{\mathrm{FMO}}$, which is the quantity that corresponds to γ_ϕ in Eq. (1). The estimate scales with γ_c^{-1}; a 100 fs relaxation time, also attributed to Ref. [38] in the literature, would double it. The optimum $\gamma_\phi^* = 1.9\,\Delta_{\mathrm{FMO}}$ is thus of the same order as the estimate (a factor 1.7 below it for 50 fs), consistent with earlier conclusions that

FMO operates near the ENAQT optimum [3, 4] and with the optimal dephasing observed in a photonic FMO simulation [27].

Rates. The size of the enhancement depends on the fast loss of the default model. With $\kappa^{-1} = 1\,\mathrm{ps}$ and $\Gamma^{-1} = 1\,\mathrm{ns}$, closer to the biological values, the coherent yield is already 0.87 and the enhancement is 1.14 ($\eta_{\mathrm{peak}} = 0.99$ at $\gamma_\phi^* = 1.78\,\Delta_{\mathrm{FMO}}$); with $\kappa^{-1} = 0.25\,\mathrm{ps}$ it is 1.04 ($\gamma_\phi^* = 1.71\,\Delta_{\mathrm{FMO}}$). The optimal rate is stable, the ratio is not; a small gain at realistic parameters is also the conclusion of Ref. [33].

Trap site. An earlier version of this work placed the trap on BChl 7; that choice gives a much larger ratio (32 against $10^{-3}\Delta_{\mathrm{FMO}}$, 39 against $\eta(0)$, at $\gamma_\phi^* = 1.55\,\Delta_{\mathrm{FMO}}$) with a similar peak yield (0.44). The difference lies in the coherent baseline: the long-time average population that coherent evolution from BChl 1 places on the trap site, without trap or loss, is 0.013 for BChl 3 but only 0.0024 for BChl 7.

F. Square lattices

As a first look beyond one dimension we computed $L \times L$ square lattices ($L = 2$–6) with uniform coupling Δ, a diagonal energy gradient from 5Δ at the source corner to 0 at the trap corner, and the same rates (80 log-spaced γ_ϕ points in $[10^{-3}, 10^{2}]\Delta$, $\eta_{\mathrm{ref}} = \eta(10^{-8}\Delta)$). Table VI lists the results. The mismatch rule carries over: each step along a shortest path lowers the energy by $\delta\varepsilon = 5\Delta/[2(L-1)]$, and γ_ϕ^* agrees with this to within 5% (2.62 versus 2.50 at $L = 2$, 0.53 versus 0.50 at $L = 6$). The enhancement stays modest (3.1–4.7) and the peak yield falls with lattice size as in 1D. For disordered lattices ($\sigma = 2\Delta$ on all sites except the trap, which stays at 0; 50 realizations per L; 50 γ_ϕ points) 76–96% of realizations have $\eta_{\mathrm{peak}}/\eta_{\mathrm{ref}} > 1.01$. Unlike the 1D criterion, this count does not require an interior maximum. Larger ensembles and other geometries are needed before drawing conclusions about dimensionality.

IV. DISCUSSION

What is robust. In ordered funnels and square lattices the optimal dephasing rate equals the nearest-neighbour energy mismatch, the HSR resonance applied to every bond, and this survives two-decade changes of the trap and loss rates. The peak yield decreases smoothly

TABLE VI. Square lattices with a diagonal gradient ($\kappa = 0.1\Delta$, $\Gamma = 0.01\Delta$), with the nearest-neighbour mismatch $\delta\varepsilon = 5\Delta/[2(L-1)]$. The last column is for disordered lattices.

L	sites	γ_ϕ^* [Δ]	$\delta\varepsilon$ [Δ]	η_{peak}	$\eta_{\text{peak}}/\eta_{\text{ref}}$	disordered, $\eta_{\text{peak}}/\eta_{\text{ref}} > 1.01$
2	4	2.62	2.50	0.686	3.05	76%
3	9	1.26	1.25	0.497	3.69	92%
4	16	0.82	0.83	0.359	4.66	96%
5	25	0.61	0.63	0.264	4.14	88%
6	36	0.53	0.50	0.200	3.92	90%

with length. Both quantities are directly comparable with experiment, where the yield and the dephasing rate are the measurable quantities [26].

What dephasing does here. Because pure dephasing is an infinite-temperature bath, it cannot favour downhill transfer, and the yield is the same for a funnel and its reverse. The flat chain, which has no ENAQT at all, transports better than every funnel, and weak random disorder with the funnel's typical step beats the funnel too. Within this model ENAQT is best read as dephasing repairing the damage done by energy mismatch. Whether a real energy landscape gains from dephasing beyond that depends on physics the model lacks, chiefly thermal relaxation toward lower sites.

What is not robust. The enhancement ratio mixes two different quantities: how well dephasing-assisted transport works, and how badly coherent transport fails. In long, strongly biased or disordered chains the second dominates. As a result the ratio (i) has no unique length dependence, since it can plateau or grow by orders of magnitude depending on the reference rate, the total bias and the loss rate; (ii) is largest for disordered chains whose absolute yield is worse; (iii) when optimized, has no finite maximum and selects landscapes that block coherent transfer; and (iv) for FMO falls from 4.7 to 1.14 when the trap and loss rates are made biological. We therefore recommend reporting η_{peak} and γ_ϕ^*, with the ratio given only together with the exact reference point and rates used.

Relation to earlier work. The mismatch rule follows from the HSR rate [7, 8] and was derived for the boundary-driven current of tilted chains, including its $1/L$ form at fixed total tilt, by Jacob *et al.* [20]; related length laws exist for disordered and lossy chains [14, 16, 21]. Our contribution is to show that the same rule fixes the optimum of the trapping yield to

within 2% for funnels of $N = 2$–20, that it survives two-decade changes of the trap and loss rates, and that it carries over to square lattices; the exponent -1.20 of a power-law fit, which we reported in earlier versions, adds nothing beyond it. The near-universal presence of ENAQT in disordered chains agrees with Refs. [8, 13–16]; our ensemble adds length-resolved statistics with the peak yield and baseline separated and a direct comparison with ordered and flat chains. We found no earlier statement that the trapping yield is invariant under bias reversal or that the flat chain beats every funnel under pure dephasing, although both follow directly from the infinite-temperature character of the bath [8, 20]. The critique of the enhancement ratio builds on earlier remarks that enhancement depends on the comparison regime [31] and on the practice of optimizing absolute flux [16, 28].

Limitations. The model is Markovian with uniform, uncorrelated pure dephasing; it neglects non-Markovian bath memory, which matters for FMO on timescales below a picosecond [10, 38], and thermal relaxation toward lower sites, which the HSR dephasing term cannot describe and which would break the bias symmetry; in this model the word funnel describes the shape of the landscape, not a preferred direction. Disorder ensembles are moderate in size (100 realizations per N for $\sigma = 2\Delta$, 50 elsewhere), and the optimizer results rest on a single run per N. Appendix A explains why we do not use the QD3SET-1 spin-boson data as a validation of the trapping model.

V. CONCLUSION

Within a minimal Lindblad model of a dephased chain with a trap:

1. The optimal dephasing rate equals the nearest-neighbour energy mismatch: $\gamma_\phi^* \approx \varepsilon$ for two sites, $\gamma_\phi^* = \varepsilon_{\mathrm{tot}}/(N-1)$ to within 2% for funnels of $N = 2$–20 sites, to within 15% across two decades of trap and loss rates, and to within 5% for square lattices.

2. The yield is invariant under bias reversal, and the flat chain beats every funnel; dephasing recovers yield lost to mismatch.

3. ENAQT occurs in 97.4% of 1,000 disordered chains ($\sigma = 2\Delta$, $N = 2$–15).

4. The enhancement ratio is dominated by the coherent baseline: its plateau in funnels, its large values in disordered chains and the unbounded values reached by optimizing it all reflect a vanishing baseline, not better transport.

5. For FMO with the trap on BChl 3 the optimal dephasing rate, $1.9\,\Delta_{\mathrm{FMO}}$, is of the order of a standard estimate of the protein dephasing rate, and is stable when the rates are made biological, while the enhancement falls to 1.14.

ACKNOWLEDGMENTS

Code, data and the scripts that regenerate every number and figure in this paper are available at https://github.com/GrandMastaShake/enaqt-open-quantum-chains. Computations used NumPy, SciPy and Matplotlib. Parts of the code were developed with AI coding assistants; the author is responsible for all physics, interpretation and text.

Appendix A: Remarks on the QD3SET-1 spin-boson data

An earlier version of this work compared the trapping model with the 1,000 numerically exact HEOM spin-boson trajectories of the QD3SET-1 dataset [39] ($\lambda = 0.1$–$1.0\,\Delta$, $\gamma_c = 1$–$10\,\Delta$, $\beta = 0.1$–$1.0\,\Delta^{-1}$, $\varepsilon \in \{0, \Delta\}$, $\rho_{11}(0) = 1$, $t \leq 20\,\Delta^{-1}$). Binning all trajectories by the high-temperature estimate $\gamma_\phi = 2\lambda/(\beta\gamma_c)$ gives a non-monotonic mean of $\rho_{22}(t = 20\,\Delta^{-1})$ for $\varepsilon = \Delta$, with a maximum of 0.771 at $\gamma_\phi \approx 0.13\Delta$ against 0.608 at the lowest bin. We no longer treat this as evidence for ENAQT or as a validation of Eq. (1), for three reasons. First, there is no trap: ρ_{22} at a fixed time measures relaxation toward thermal equilibrium, not a trapping yield. Second, the binned curve pools trajectories with different temperatures and reorganization energies, so its shape partly reflects how the parameter grid maps onto γ_ϕ. Third, the HEOM bath includes thermal relaxation and memory that the pure dephasing model lacks. A quantitative test of the trapping model against exact dynamics would need HEOM (or similar) simulations that include the trap, which we leave for future work.

Appendix B: Reproducibility

`scripts/paper_numbers.py` recomputes every number in Secs. III A–III C and III E, the step profiles and baselines of Sec. III D, and the mismatch comparison of Sec. III F (disorder seeds 0–99 with `numpy.random.default_rng`, bootstrap seed 0; about 80 s on one core) and writes `paper_numbers.json`; `scripts/make_paper_figures.py` draws Figs. 1–4 from that

file. The optimized site energies of Sec. III D come from `enaqt_optimal_disorder.py`, the square-lattice runs of Sec. III F from `enaqt_2d_lattice.py`, and the QD3SET-1 analysis of Appendix A from `enaqt_sb_analysis.py`; these scripts take longer and their outputs are stored as JSON in the repository. The test suite (`tests/`) pins the key numbers and checks the mismatch rule, the bias symmetry and the flat-chain results.

---

[1] M. Mohseni, P. Rebentrost, S. Lloyd, and A. Aspuru-Guzik, Environment-assisted quantum walks in photosynthetic energy transfer, J. Chem. Phys. **129**, 174106 (2008).

[2] M. B. Plenio and S. F. Huelga, Dephasing-assisted transport: quantum networks and biomolecules, New J. Phys. **10**, 113019 (2008).

[3] P. Rebentrost, M. Mohseni, I. Kassal, S. Lloyd, and A. Aspuru-Guzik, Environment-assisted quantum transport, New J. Phys. **11**, 033003 (2009).

[4] F. Caruso, A. W. Chin, A. Datta, S. F. Huelga, and M. B. Plenio, Highly efficient energy excitation transfer in light-harvesting complexes: The fundamental role of noise-assisted transport, J. Chem. Phys. **131**, 105106 (2009).

[5] P. W. Anderson, Absence of diffusion in certain random lattices, Phys. Rev. **109**, 1492 (1958).

[6] H. Haken and P. Reineker, The coupled coherent and incoherent motion of excitons and its influence on the line shape of optical absorption, Z. Phys. **249**, 253 (1972).

[7] H. Haken and G. Strobl, An exactly solvable model for coherent and incoherent exciton motion, Z. Phys. **262**, 135 (1973).

[8] J. M. Moix, M. Khasin, and J. Cao, Coherent quantum transport in disordered systems: I. The influence of dephasing on the transport properties and absorption spectra on one-dimensional systems, New J. Phys. **15**, 085010 (2013).

[9] J. Wu, F. Liu, Y. Shen, J. Cao, and R. J. Silbey, Efficient energy transfer in light-harvesting systems, I: optimal temperature, reorganization energy and spatial-temporal correlations, New J. Phys. **12**, 105012 (2010).

[10] J. Moix, J. Wu, P. Huo, D. Coker, and J. Cao, Efficient energy transfer in light-harvesting systems, III: The influence of the eighth bacteriochlorophyll on the dynamics and efficiency in FMO, J. Phys. Chem. Lett. **2**, 3045 (2011).

[11] I. Kassal and A. Aspuru-Guzik, Environment-assisted quantum transport in ordered systems, New J. Phys. **14**, 053041 (2012).

[12] M. Žnidarič and M. Horvat, Transport in a disordered tight-binding chain with dephasing, Eur. Phys. J. B **86**, 67 (2013).

[13] L. Novo, M. Mohseni, and Y. Omar, Disorder-assisted quantum transport in suboptimal decoherence regimes, Sci. Rep. **6**, 18142 (2016).

[14] Y. Zhang, G. L. Celardo, F. Borgonovi, and L. Kaplan, Optimal dephasing for ballistic energy transfer in disordered linear chains, Phys. Rev. E **96**, 052103 (2017).

[15] E. Zerah-Harush and Y. Dubi, Effects of disorder and interactions in environment assisted quantum transport, Phys. Rev. Research **2**, 023294 (2020).

[16] A. R. Coates, B. W. Lovett, and E. M. Gauger, Localisation determines the optimal noise rate for quantum transport, New J. Phys. **23**, 123014 (2021).

[17] E. Zerah-Harush and Y. Dubi, Universal origin for environment-assisted quantum transport in exciton transfer networks, J. Phys. Chem. Lett. **9**, 1689 (2018).

[18] A. R. Coates, B. W. Lovett, and E. M. Gauger, From Goldilocks to twin peaks: multiple optimal regimes for quantum transport in disordered networks, Phys. Chem. Chem. Phys. **25**, 10103 (2023).

[19] S. Burkhardt, M. Kraft, R. Mannella, and S. Wimberger, Noise-assisted transport in the Wannier-Stark system, New J. Phys. **15**, 045008 (2013).

[20] S. L. Jacob, L. P. Bettmann, A. M. Lacerda, K. Zawadzki, S. R. Clark, J. Goold, and J. J. Mendoza-Arenas, Dephasing-assisted transport in a tight-binding chain with a linear potential, Front. Phys. **12**, 1474018 (2024).

[21] A. Skalkin, R. Unanyan, and M. Fleischhauer, Dephasing enhanced transport of spin excitations in a two dimensional lossy lattice, arXiv:2502.10854 (2025).

[22] H. Yeler and Ö. Çakır, Environment-assisted transport in a strongly correlated boundary-driven Fermi-Hubbard chain, arXiv:2609.05137 (2026).

[23] S. Viciani, M. Lima, M. Bellini, and F. Caruso, Observation of noise-assisted transport in an all-optical cavity-based network, Phys. Rev. Lett. **115**, 083601 (2015).

[24] D. N. Biggerstaff, R. Heilmann, A. A. Zecevik, M. Gräfe, M. A. Broome, A. Fedrizzi, S. Nolte, A. Szameit, A. G. White, and I. Kassal, Enhancing coherent transport in a photonic network using controllable decoherence, Nat. Commun. **7**, 11282 (2016).

[25] A. Potočnik, A. Bargerbos, F. A. Y. N. Schröder, S. A. Khan, M. C. Collodo, S. Gasparinetti, Y. Salathé, C. Creatore, C. Eichler, H. E. Türeci, A. W. Chin, and A. Wallraff, Studying light-harvesting models with superconducting circuits, Nat. Commun. **9**, 904 (2018).

[26] C. Maier, T. Brydges, P. Jurcevic, N. Trautmann, C. Hempel, B. P. Lanyon, P. Hauke, R. Blatt, and C. F. Roos, Environment-assisted quantum transport in a 10-qubit network, Phys. Rev. Lett. **122**, 050501 (2019).

[27] H. Tang *et al.*, Simulating photosynthetic energy transport on a photonic network, npj Quantum Inf. **10**, 29 (2024).

[28] M. Lawrence, M. Pocrnic, E. Fung, J. Carrasquilla, E. M. Gauger, and D. Segal, Discovery of energy landscapes toward optimized quantum transport: Environmental effects and long-range tunneling, Phys. Rev. Research **8**, 013163 (2026).

[29] M. Lawrence, E. Wang, and D. Segal, Design principles for enhanced quantum transport with site-dependent noise, Phys. Rev. B **114**, 034210 (2026).

[30] L. Gassab, B. Adams, Y. H. Goolam Hossen, A. Hassasfar, O. Pusuluk, N. J. Murugan, I. K. Kominis, Ö. E. Müstecaplıoğlu, F. Petruccione, and T. J. A. Craddock, Quantum in biology, quantum for biology, and biology for quantum: mapping the evidence and the road ahead, arXiv:2605.00205 (2026).

[31] I. Kassal, J. Yuen-Zhou, and S. Rahimi-Keshari, Does coherence enhance transport in photosynthesis?, J. Phys. Chem. Lett. **4**, 362 (2013).

[32] D. Dwiputra and F. P. Zen, Environment-assisted quantum transport and mobility edges, arXiv:2012.09337 (2020).

[33] E. Zerah Harush and Y. Dubi, Do photosynthetic complexes use quantum coherence to increase their efficiency? Probably not, Sci. Adv. **7**, eabc4631 (2021).

[34] J. J. Mendoza-Arenas and S. R. Clark, Giant rectification in strongly interacting driven tilted systems, PRX Quantum **5**, 010341 (2024).

[35] G. H. Wannier, Wave functions and effective Hamiltonian for Bloch electrons in an electric field, Phys. Rev. **117**, 432 (1960).

[36] M. Kappus and F. Wegner, Anomaly in the band centre of the one-dimensional Anderson model, Z. Phys. B **45**, 15 (1981).

[37] J. Adolphs and T. Renger, How proteins trigger excitation energy transfer in the FMO complex of green sulfur bacteria, Biophys. J. **91**, 2778 (2006).

[38] A. Ishizaki and G. R. Fleming, Theoretical examination of quantum coherence in a photosynthetic system at physiological temperature, Proc. Natl. Acad. Sci. USA **106**, 17255 (2009).

[39] A. Ullah, L. E. Herrera Rodriguez, P. O. Dral, and A. A. Kananenka, QD3SET-1: a database with quantum dissipative dynamics datasets, Front. Phys. **11**, 1223973 (2023).